



```
In [1]: !pip install pyperclip
```

Requirement already satisfied: pyperclip in /usr/local/lib/python3.12/dist-packages (1.11.0)

```
In [6]: import random, string
import pyperclip

def generate_password(length):
    password = []

    # Ensure at least one character from each type if length is sufficient
    if length >= 4:
        password.append(random.choice(string.ascii_uppercase))
        password.append(random.choice(string.ascii_lowercase))
        password.append(random.choice(string.digits))
        password.append(random.choice(string.punctuation))

        # Fill the rest with random choices until the specified length
        for _ in range(length - 4):
            password.append(random.choice(string.ascii_uppercase + string.ascii_lowercase + string.digits + string.punctuation))

        # Shuffle to ensure randomness
        random.shuffle(password)
    else:
        # If length is less than 4, just fill the required length with random
        for _ in range(length):
            password.append(random.choice(string.ascii_uppercase + string.ascii_lowercase + string.digits + string.punctuation))

    return ''.join(password)

# --- Non-GUI interaction for Colab ---
# Get password length from user input
while True:
    try:
        pass_length = int(input("Enter password length (8-32): "))
        if 8 <= pass_length <= 32:
            break
        else:
            print("Length must be between 8 and 32.")
    except ValueError:
        print("Invalid input. Please enter a number.")

# Generate and display the password
new_password = generate_password(pass_length)
print(f"Generated Password: {new_password}")

# Optionally copy to clipboard (if pyperclip works in the environment)
try:
    pyperclip.copy(new_password)
    print("Password copied to clipboard.")
except pyperclip.PyperclipException:
    print("Could not copy to clipboard. Pyperclip might not be fully supported")
```

Enter password length (8-32): sadhu prasanna lakshmi jyothi

Invalid input. Please enter a number.

Enter password length (8-32): Prasanna

Invalid input. Please enter a number.

Enter password length (8-32): 10

Generated Password: ,}G#w0<6E4

Could not copy to clipboard. Pyperclip might not be fully supported in this environment.